Artemy Urodovskikh

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github.com/R2BPW · r2bpw.works

Education

MSc in Computer Science *2022*
Moscow Institute of Physics and Technology (MIPT).
Thesis: *Visualization algorithms for large-scale route data on maps.*

BSc in Software Engineering *2020*
MIREA – Russian Technological University.

Professional Experience

Backend Team Lead *May 2018 – present*
Bolshaya Troika, Moscow. 8+ years.

Core product: regional-scale waste-management platform (Python/Django, PostgreSQL, ClickHouse) for government clients in Russia and Kazakhstan. Led a team of five engineers within a 40-developer codebase. Owned releases, rollbacks, and architecture decisions for the core platform.

Route Planning (core owner)

- Developed the route-planning subsystem from scratch: background task queue for route calculations, synchronous solver for 10,000+ waypoints per district, 500+ per vehicle.
- Implemented a standalone TSP solver with graceful degradation. In production since early 2022.

Billing Engine (core owner)

- Designed and developed the billing system from scratch: tariff calculation, accruals, invoice generation for 1M+ individual payers (municipal waste collection fees).
- Memory-optimized batch printing system. Led cross-cloud storage migration.

Fleet Telemetry (contributor)

- Data synchronization layer between ClickHouse and PostGIS.
- Designed and implemented GPS filtering heuristics (track splitting, segment merging, outlier rejection) to recover actual movement geometry from drifting raw GPS streams.
- Integrated S3 storage for onboard camera footage.

Statistics and Analytics (core owner)

- Analytical subsystem from domain model to dashboards: scheduled ETL, geospatial clustering (DBSCAN), two-stage weighing reconciliation, ClickHouse for regional-scale queries.

Object Versioning (core owner)

- Parallel versioning strategy: multiple concurrent drafts with change-set merging on publish. Delta storage; relationship copying logic configurable through admin without code changes.

Performance Engineering

- Non-blocking index creation on production tables with millions of rows. Eliminated N+1 queries via ORM annotations. Replaced a nested-loop plan on a 280M-row table with a hash anti-join, resolving statement timeouts.

Logistics Algorithms and Visualization

- Large-scale geospatial data: vehicle tracks, nested territories, graph overlays at regional scale.

Hardware–Software Complex for Municipal Inventory

- Automated yard territory inventory system. Still serving original clients.

Teaching

- Teaching assistant, MUCTR (2020); MIPT (2022).
- Code reviewer and mentor, Yandex.Practicum, backend development track (2023–2024).

Technical Skills

Languages	Python, SQL, C
Backend	Django, Django REST Framework, Celery, PostgreSQL, PostGIS, ClickHouse, Redis
Infrastructure	Docker, Kubernetes, Terraform, GitLab CI, S3 (Yandex/MinIO), Nginx, Grafana
Domain Expertise	GIS, route optimization (TSP/CVRP), telemetry, billing, DBSCAN clustering, EAV/versioning
Other	FPGA (Xilinx Spartan-6), SDR, IrDA, Git, pytest, Agile

Certifications and Courses

- Cisco CCNA (2019).
- Practical Reinforcement Learning – HSE, Coursera (2021).
- Big Data Essentials: HDFS, MapReduce and Spark RDD – Yandex, Coursera (2020).

Selected Projects

Radio Bargain Bot *r2bpw.works*
Price-intelligence system for the secondary ham radio market. Aggregates listings from 98 sources across 50+ countries, scores each deal using Bayesian bootstrap benchmarks, publishes verified bargains to 12 regional Telegram channels.

Custom SDR Receiver on FPGA *GitHub*
SDR receiver firmware for Xilinx Spartan-6 (OSA103 Mini), written from scratch in Verilog: NCO, hardware-multiplier mixer, decimating CIC filter.

IR Bridge – Ericsson MC218 ↔ Flipper Zero ↔ LLM *r2bpw.works*
Bidirectional infrared bridge from a 1999 Ericsson MC218 palmtop to a large language model. Flipper Zero decodes IrDA and relays via ESP32 to an LLM API.

30m QRSS Beacons *r2bpw.works*
Catalogue of identified QRSS beacons on the 30-metre band, built on a distributed network of KiwiSDR receivers worldwide.